

JACOB RENN

jacobrenn.github.io ◇ github.com/jacobrenn ◇ linkedin.com/in/jacobrenn-aisquared

1261 Pekin Road ◇ Pasadena, MD 21122

(301) 514-5057 ◇ jacob.renn4@gmail.com

PROFESSIONAL SUMMARY

AI Researcher, Engineer, and Architect with a PhD in Explainable AI, dedicated to bridging the gap between fundamental AI research and production-scale deployment. Holds a TS/SCI clearance with Full-Scope polygraph. Proven track record of delivering high-impact AI capabilities for the DoD and Intelligence Community, serving as Principal Investigator (PI) on multiple SBIR/STTR and CRADA engagements. Architected and led the development of multiple families of LLMs and currently serves as Co-Founder and Chief Data Scientist at AISquared, specializing in the development and deployment of robust AI capabilities.

SECURITY CLEARANCE

Top Secret/SCI with Full-Scope Polygraph

EXPERIENCE

Chief Data Scientist and Co-Founder, AI Squared, Inc.

September 2020 - Present

- Architected and led the end-to-end development of the Bolt LLM family, including choosing base models, designing and implementing training data, and training models optimized for complex cognitive behaviors including enhanced guardrail implementations, domain-specific fine-tuning, and efficiency for on-premise deployments.
- Engineered a modular AI service layer within the AISquared UNIFI platform, developing a suite of production APIs that operationalize hosted models into specialized engines, including a robust guardrails engine, a model routing system, an LLM-as-a-judge framework, and high-throughput services for embeddings, translation, and general-purpose chat.
- Designed and implemented a highly-capable agentic harness enabling LLMs to operate in a closed-loop, tool-use environment; engineered a fully-customizable orchestration layer supporting dynamic tool integration, skill-extension frameworks, and system prompt configuration to ensure reliable, multi-step task execution and security.
- Served as Principal Investigator (PI) on multiple SBIR/STTR and CRADA engagements, securing non-dilutive funding and directing the technical delivery of AI capabilities for various Intelligence Community (IC) and DoD customers.
- Trained high-precision classification models for LLM guardrails, safety protocols, and model routing; significantly mitigated malicious or misaligned queries and enforced strict security postures across multi-agent architectures to meet rigorous deployment requirements.
- Led the design and implementation of a browser-based AI integration technology stack and an open-source Python SDK, solving the “last mile of machine learning” by enabling seamless model integration into existing mission workflows.
- Authored and contributed the BeyondML project to the Linux Foundation Data and AI, introducing a novel architecture that enables a single neural network to perform multiple tasks via isolation of network weights.

Adjunct Faculty, Capitol Technology University

May 2022 - April 2026

- Directed doctoral research in generative AI and transformer architectures, overseeing experimentation and methodology for PhD candidates.
- Served as an external examiner for doctoral dissertations, evaluating research rigor and academic contribution to ensure successful PhD completion.
- Led instruction for *CS-150: Programming in C*, achieving an overall student excellence rating of 4.6/5.

Delivery Data Scientist, Microsoft Corporation

May 2020 - November 2021

- Led data science delivery for Microsoft’s Intelligence Community clients, bridging the gap between enterprise cloud infrastructure and specialized AI mission needs.
- Developed and productionalized custom ML models for government PoCs, enabling advanced predictive analytics for national security workflows, including custom models for malware classification based on executable bytes and content.
- Supported contract bids by assessing requirements and proposing technical solutions for federal agencies.
- Collaborated with teams to modernize system architecture, improving data processing efficiency for key systems providing enterprise-wide data processing capabilities.

Data Scientist, WaveStrike, LLC*June 2019 - April 2020*

- Engineered high-performance text analytics software supporting the training and evaluation of NLP models.
- Optimized transformer-based architectures (BERT, RoBERTa) for few-shot classification tasks, leveraging High-Performance Computing (HPC) environments to maximize training efficiency and model performance.
- Spearheaded the technical development of research proposals for the NSF and DoD to develop novel interpretable AI technologies.

Data Scientist, United States Department of Defense*July 2018 - June 2019*

- Developed and deployed a document-matching tool using enterprise data repositories and latent semantic indexing to enable analysts to match documented information needs to draft reports.
- Researched and applied natural language processing techniques for topic modeling and document similarity analysis.
- Engineered data pipelines and automation workflows, reducing repeated task man-hours by approximately 50%.
- Developed image classification models and custom Python packages to meet project-specific needs.

Research Assistant, University of Maryland, College Park*May 2018 - December 2018*

- Designed and tested a novel decision tree algorithm, achieving superior performance on complex datasets compared to industry-standard decision tree models.
- Published research findings internally and contributed to academic discussions on machine learning.

Teaching Assistant, University of Maryland, College Park*August 2017 - May 2018*

- Taught and graded undergraduate calculus courses, providing additional support through office hours.

EDUCATION

Doctor of Philosophy in Technology, Capitol Technology University*2022*

- Research Area: Explainable Artificial Intelligence
- Dissertation: *Linear Regression Feature Engineering in Classification Tree Learning*

Master of Science in Business Analytics, University of Maryland*2018*

- GPA: 4.0

Bachelor of Science in Mathematics, University of Maryland*2017*

- GPA: 3.77, Honors College Citation

TECHNICAL SKILLS

- **AI/ML:** Large Language Models (LLMs), Explainable AI (XAI), Agentic Frameworks, Fine-tuning (PEFT, LoRA), RLHF, BERT, RoBERTa, Latent Semantic Indexing, Topic Modeling.
- **LLM Engineering:** Quantization (GGUF, bitsandbytes), vLLM, TGI, Vector Databases, Advanced Chunking Strategies, RAG Architecture.
- **Frameworks & Libraries:** PyTorch, Transformers (HuggingFace), FastAPI, Dask, Scikit-learn, Pandas, NumPy.
- **Infrastructure & MLOps:** Docker, MLflow, Weights & Biases, GitHub Actions (CI/CD), DeepSpeed, Slurm, RabbitMQ, On-premise Deployment, HPC Environments, GPU Acceleration.
- **Languages:** Python, JavaScript, C, LaTeX.
- **Specializations:** Interpretability and Explainability (XAI), Tool-use Orchestration, Production RAG, LLM-as-a-judge, Efficient Inference Optimization, Guardrail Implementation, Model Routing.